

# QUIC PROTOCOL ANALYSIS REPORT

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## OVERVIEW

QUIC is a

- secure general-purpose transport protocol.
- connection-oriented protocol – creates stateful interaction between server and client

**Quick Handshake** combines negotiation of **Cryptographic** + **Transport** parameters.

Includes option for clients to send data immediately (**0-RTT**) (prior communication/configuration required)

Application protocols exchange information over a QUIC Connection via **streams (Ordered sequence of bytes)**.

QUIC depends on congestion control to avoid network congestion.

QUIC connections are **not strictly bounded** to a single network path. Connection migration uses **Connection IDs** to allow connections to transfer to a new network path (**Only clients** can migrate in Version 1)

**Current versions:**

- Version 1 (RFC 9000): 0x00000001 ( **Version-Independent Properties** )
- Version 2 (RFC 9369): 0x6b3343cf

## STREAMS

Application protocols exchange information over a QUIC Connection via **streams (Ordered sequence of bytes)**. They last the **entire** duration of the connection.

Two types of streams:

- i) **Bidirectional streams**: Allow both endpoints to send data
- ii) **Unidirectional streams**: Allow a single endpoint to send data

Identified by a numeric value, called **Stream ID** (62-bit integer), unique for all streams on a connection. They are encoded as Variable-Length Integers. (Must NOT be reused within a connection).

- **LSB** of Stream ID is the **initiator** (0 – Client, 1 – Server)
- **Second LSB** of Stream ID identifies **Unidirectional** (1) / **Bidirectional** (0) Stream.

# CONNECTION

Connection starts with **Handshake Phase**:

- The two endpoints established a shared secret key using the cryptographic handshake protocol.
- This handshake confirms that both endpoints are willing to communicate and establishes parameters for connection.

**CLIENT -> Initial [0]: CRYPTO (Client HELLO)-> SERVER**

**SERVER -> Initial [0]: CRYPTO (Server HELLO) & Handshake [0]: CRYPTO -> CLIENT**

**CLIENT -> Handshake [1]: CRYPTO (Finished) -> SERVER**

**SERVER -> 1-RTT Data (Encrypted) -> CLIENT**

An application protocol can use the connection during the handshake phase with some limitations:

- RTT allows application data to be sent by a client before receiving a response from a server. However, 0-RTT does **NOT** provide any protection against replay attacks.
- A server can also send application data to a client before it receives the final cryptographic handshake messages that allow it to confirm the liveness and identity of the client.

**These capabilities allow the application protocol to offer the option of trading some security guarantees for reduced latency.**

**Connections IDs** allow connections to migrate to a new network path, both by choice of an endpoint, and when forced to change in the middlebox.

**Primary function:** To ensure that the changes happening at the lower protocol layers (UDP, IP) does not cause the QUIC packets to be delivered to the wrong endpoint.

# PACKET FORMATS

All numeric values are encoded in network byte order, and all field sizes are in bits (in hexadecimal notation).

Packet Numbers: **0 –  $2^{62}$  -1 (62-bit)**

**Encoding:** In long or short packet headers, they are encoded in 1 to 4 bytes. Only least significant bits of the packet number are included to reduce the size. This is then encrypted.

**Decoding:** At receiver end, first the protection of the packet number is removed. The full packet number is reconstructed based on:

- I. No. of significant bits present
- II. Largest packet number received

- i) **Long Header Packets: Source Connection ID + Destination Connection ID (for new connections)** – Allows special packets (Version Negotiation packet)

```
Long Header Packet {
  Header Form (1) = 1,
  Fixed Bit (1) = 1,
  Long Packet Type (2),
  Type-Specific Bits (4),
  Version (32),
  Destination Connection ID Length (8),
  Destination Connection ID (0..160),
  Source Connection ID Length (8),
  Source Connection ID (0..160),
  Type-Specific Payload (...),
}
```

Figure 13: Long Header Packet Format

- ii) **Short Header Packets: Only Destination Connection ID**

- **Long Headers** are used for packets sent prior to establishment of 1-RTT keys. Once 1-RTT keys are available, the sender switches to **short header**.
- Once header protection is removed, the packet number is decoded by finding the packet number value closest to the next expected packet (Highest received packet no. + 1).

## VERSION NEGOTIATION

Allows server to indicate that it does not support the version the client used, by sending a Version Negotiation Packet to any packet that might initiate a new connection.

Determined by the **first packet** sent by a client.

For clients that support multiple QUIC versions, they must ensure that the first UDP datagram sent is sized: **LARGEST( MINIMUM ( Datagram sizes supported for each version ) )**

The Version Negotiation packets includes a list of versions accepted by the server.

```
Version Negotiation Packet {
  Header Form (1) = 1,
  Unused (7),
  Version (32) = 0,
  Destination Connection ID Length (8),
  Destination Connection ID (0..2040),
  Source Connection ID Length (8),
  Source Connection ID (0..2040),
  Supported Version (32) ...,
}
```

Figure 14: Version Negotiation Packet

# QUIC vs Generic UDP

## 1. Packet Header Structure Differences

**Generic UDP (42 Bytes):** Simple

- Ethernet Header (14 bytes)
- IP Header (20 bytes for IPv4)
- UDP Header (8 bytes)
- Payload (Variable Length)

**QUIC:**

- Ethernet Header(14 bytes)
- IP Header (20 bytes for IPv4)
- UDP Header (8 bytes)
- QUIC Header (10-50+ bytes)
- QUIC Payload (Encrypted)

**Key Difference:** QUIC has a substantial header with identifiable structure

## 2. QUIC Header Patterns

First Byte (QUIC):

- Header Form (H): 0 = short, 1 = long
- Fixed Bit (F): 1 for QUIC
- Type: Packet Type
- Reserved (R)
- Packet Number Length (P)

First Byte (Generic UDP): Any value, no consistent pattern

Version Field (Long Header Only):

- 0x00000001 – QUIC v1
- 0x6b3343cf – QUIC v2
- 0x00000000 – Version Negotiation

Version Field (Generic UDP): ABSENT

Connection ID: Variable Length (0-20 bytes), absent in Generic UDP

### 3. Packet Size Distributions

#### QUIC Traffic Patterns

##### A. QUIC Handshake:

Packet 1 (Initial – Client Hello): 1200-1280 bytes

Packet 2 (Initial – ServerHello): 1200-1280 bytes

Packet 3-4 (Handshake): 1000-1200 bytes

Packet 5+ (1-RTT Data): Highly Variable Size (200-1400 bytes)

##### Observable pattern:

1. Initial burst of large packets (~1200 bytes)
2. Then variable-sized data packets

##### B. QUIC Web Browsing:

Typical size distribution:

- Small (< 300 bytes): (25%)
- Medium (300-800 bytes): (35%)
- Large (800-1400 bytes): (40%)

**Pattern:** Mixed sizes, bursty

**Reason:** HTTP requests are small, but responses and downloads are large

##### C. QUIC Streaming

Typical size distribution:

- Small (< 300 bytes): (5%)
- Medium (300-800 bytes): (10%)
- Large (800-1400 bytes): (85%)

**Pattern:** Dominated by large packets, steady stream

**Reason:** In video chunks, almost all packets are full sized, steady transmission rate

#### Generic UDP Traffic Patterns

##### A. DNS (Port 53)

- Query packets: 40-100 bytes
- Response packets: 50-512 bytes

**Pattern:** Small, paired query-response

#### **B. NTP (Port 123)**

All packets: EXACTLY 90 bytes (IP+UDP+NTP)

**Pattern:** Fixed size, periodic

#### **C. VoIP/RTP (Various Ports)**

All packets: ~200-220 bytes (very consistent)

**Pattern:** Constant rate, fixed size

#### **D. Online Gaming (Various Ports)**

Tiny packets: 40-150 bytes

**Pattern:** Very small, high frequency

### **4. Timing Patterns (Inter-Arrival Times)**

#### **QUIC Traffic Timing**

##### **A. QUIC Web:**

Time between packets (milliseconds):

Handshake phase:

- Packet 1 → Packet 2: ~50ms (RTT)
- Packet 2 → Packet 3: ~1ms (burst)
- Packet 3 → Packet 4: ~1ms (burst)

Data phase:

- Highly variable: 0.1ms - 100ms

**Bursty pattern:** groups of packets close together

##### **B. QUIC Streaming:**

Time between packets:

Steady stream: ~5-20ms between packets

Occasional bursts: 0.1-1ms (during buffering)

Very consistent delivery rate

## Generic UDP Timing

### A. DNS:

- Query at  $t=0$
- Response at  $t=50\text{ms}$
- [Silence for minutes]
- Next query at  $t=300000\text{ms}$

**Pattern:** Query-response pairs separated by long silence

### B. NTP:

- Request at  $t=0$
- Response at  $t=50\text{ms}$
- [Silence for  $\sim 5$  minutes]
- Next request at  $t=300000\text{ms}$

**Pattern:** Periodic, widely spaced

### C. VoIP:

- Packets every 20ms exactly

**Pattern:** Clock-like precision, constant rate

## 5. Port Usage Patterns

### QUIC

Primary ports:

- UDP 443 (HTTPS) 95%
- UDP 80 (HTTP) 3%
- UDP 8080, 4433, etc. 2%

**Pattern:** Strongly concentrated on port 443

### Generic UDP

Common ports:

- UDP 53 (DNS): Frequent
- UDP 123 (NTP): Periodic
- UDP 67/68 (DHCP): Occasional
- UDP 161 (SNMP): Monitoring
- UDP 5060 (SIP): VoIP

- UDP 3478 (STUN): WebRTC
- Random high ports: Gaming, P2P

**Pattern:** Widely distributed across port range

**Key difference:**

- QUIC: Almost always port 443
- Generic UDP: Spread across many ports



# Environment Setup Guide

Here are the steps to set up the IETF QUIC protocol stack, traffic generation tools, and packet capture permissions.

## 1. Project Directory Structure

To maintain consistency across scripts and ML models, all team members should use the following layout:

Plaintext

QUIC-Project/

```
├── libs/      # Compiled QUIC implementations (ngtcp2, quiche)
├── data/      # Raw PCAPs and SSL Keylogs (Add to .gitignore)
├── scripts/   # Python automation and ML code
└── docs/     # Project documentation
```

## 2. System Dependencies

Install the necessary compilers and networking libraries for your specific OS:

- **macOS:** brew install cmake ninja rust libev openssl@3 pkg-config
- **Linux/WSL2:** sudo apt update && sudo apt install -y cmake ninja-build rustc libev-dev libssl-dev pkg-config

## 3. Building the Protocol Stack

Using **ngtcp2** as our primary reference for IETF QUIC traffic.

### A. Build nghttp3 (The HTTP/3 Layer)

Bash

```
cd QUIC-Project/libs
git clone --recursive https://github.com/ngtcp2/nghttp3
cd nghttp3
autoreconf -i
./configure --prefix=$PWD/build --enable-lib-only
make -j$(nproc 2>/dev/null || sysctl -n hw.ncpu) install
```

### B. Build ngtcp2 (The QUIC Layer)

Bash

```
cd ../
git clone --recursive https://github.com/ngtcp2/ngtcp2
cd ngtcp2
autoreconf -i
```

# For Linux/WSL:

```
PKG_CONFIG_PATH=$PWD/../nghttp3/build/lib/pkgconfig ./configure --with-openssl
```

```
# For macOS (Apple Silicon):
./configure \
  LIBEV_CFLAGS="-I/opt/homebrew/include" \
  LIBEV_LIBS="-L/opt/homebrew/lib -lev" \
  PKG_CONFIG_PATH="$PWD/../nghttp3/build/lib/pkgconfig" \
  --with-openssl

make -j$(nproc 2>/dev/null || sysctl -n hw.ncpu)
```

## 4. Verification & Key Logging

The goal of this project is to identify traffic. To verify your model's accuracy, you must be able to decrypt the traffic using an **SSLKEYLOGFILE**.

### The Smoke Test

Run the following commands to confirm your client can connect to the internet and log its secrets:

1. **Create the data directory:**  
`mkdir -p ~/QUIC-Project/data`
2. **Run the client:**  
Bash  
`export SSLKEYLOGFILE=~/QUIC-Project/data/keys.log`  
`./examples/opensslclient cloudflare-quic.com 443`
3. **Confirm the Log:**  
`cat ~/QUIC-Project/data/keys.log`  
(You should see lines starting with `CLIENT_HANDSHAKE_TRAFFIC_SECRET`)

## 5. Packet Capture Permissions

To allow Python scripts to capture traffic without root access:

- **macOS:** Open Wireshark → Menu → **Install ChmodBPF**.
- **Linux:** Run `sudo usermod -aG wireshark $USER` then log out and back in.

## Troubleshooting Tips

- **Binary Names:** On macOS/Linux, the client is usually located at `libs/ngtcp2/examples/opensslclient`.
- **MTU Issues:** If you see `sendmsg: Message too long`, it is normal. QUIC is performing Path MTU Discovery (PMTUD).

# TRAFFIC CAPTURE

## QUIC DATA

(using Chrome QUIC) (quic\_google\_01.pcap) (chrome\_keys.log)

|    |          |                |                 |       |      |                                                                               |
|----|----------|----------------|-----------------|-------|------|-------------------------------------------------------------------------------|
| 27 | 1.620854 | 192.178.211.84 | 172.17.93.192   | QUIC  | 82   | Initial, SCID=e36060b6d55e224e, PKN: 1, ACK                                   |
| 28 | 1.621683 | 192.178.211.84 | 172.17.93.192   | QUIC  | 1292 | Initial, SCID=e36060b6d55e224e, PKN: 2, ACK, PADDING                          |
| 29 | 1.621684 | 192.178.211.84 | 172.17.93.192   | QUIC  | 1292 | Initial, SCID=e36060b6d55e224e, PKN: 3, CRYPTO, PADDING                       |
| 30 | 1.622426 | 192.178.211.84 | 172.17.93.192   | QUIC  | 1292 | Initial, SCID=e36060b6d55e224e, PKN: 4, CRYPTO, PADDING                       |
| 31 | 1.623359 | 192.178.211.84 | 172.17.93.192   | QUIC  | 1292 | Handshake, SCID=e36060b6d55e224e, PKN: 5, CRYPTO                              |
| 32 | 1.623697 | 172.17.93.192  | 192.178.211.84  | QUIC  | 1292 | Initial, DCID=e36060b6d55e224e, PKN: 3, ACK, PADDING                          |
| 33 | 1.624454 | 192.178.211.84 | 172.17.93.192   | QUIC  | 1292 | Handshake, SCID=e36060b6d55e224e, PKN: 7, CRYPTO                              |
| 34 | 1.624648 | 172.17.93.192  | 192.178.211.84  | QUIC  | 81   | Handshake, DCID=e36060b6d55e224e, PKN: 4, ACK                                 |
| 35 | 1.625296 | 192.178.211.84 | 172.17.93.192   | QUIC  | 1292 | Handshake, SCID=e36060b6d55e224e, PKN: 5, CRYPTO                              |
| 36 | 1.625476 | 172.17.93.192  | 192.178.211.84  | QUIC  | 82   | Handshake, DCID=e36060b6d55e224e, PKN: 5, ACK                                 |
| 37 | 1.648877 | 172.17.93.192  | 142.251.221.164 | QUIC  | 125  | Handshake, DCID=e49cf2145a03ecd2, PKN: 11, CRYPTO                             |
| 38 | 1.648931 | 172.17.93.192  | 142.251.221.164 | HTTP3 | 109  | SETTINGS                                                                      |
| 40 | 1.649331 | 172.17.93.192  | 142.251.221.164 | HTTP3 | 1288 | PRIORITY_UPDATE, STREAM(10), QPACK_ENC[23]                                    |
| 41 | 1.649333 | 172.17.93.192  | 142.251.221.164 | HTTP3 | 926  | QPACK_ENC[61], STREAM(0), HEADERS: GET https://www.google.com/async/fo/ae?asy |
| 42 | 1.651536 | 172.17.93.192  | 192.178.211.84  | QUIC  | 79   | Handshake, DCID=e36060b6d55e224e, PKN: 7, PADDING, PING                       |
| 43 | 1.653521 | 192.178.211.84 | 172.17.93.192   | HTTP3 | 844  | SETTINGS                                                                      |
| 44 | 1.656582 | 172.17.93.192  | 192.178.211.84  | QUIC  | 82   | Handshake, DCID=e36060b6d55e224e, PKN: 8, ACK                                 |
| 45 | 1.657710 | 172.17.93.192  | 192.178.211.84  | QUIC  | 125  | Handshake, DCID=e36060b6d55e224e, PKN: 9, CRYPTO                              |
| 46 | 1.657790 | 172.17.93.192  | 192.178.211.84  | HTTP3 | 116  | SETTINGS                                                                      |
| 47 | 1.658484 | 172.17.93.192  | 192.178.211.84  | HTTP3 | 1288 | PRIORITY_UPDATE, STREAM(10), QPACK_ENC[25]                                    |
| 48 | 1.658489 | 172.17.93.192  | 192.178.211.84  | HTTP3 | 1292 | QPACK_ENC[61]                                                                 |

> Frame 7: Packet, 1292 bytes on wire (10336 bits), 1292 bytes captured (10336 bits) on interface 0

> Ethernet II, Src: Apple\_a8:84:58 (68:ca:c4:a8:84:58), Dst: Hewlett-Packard\_a8:84:1a (68:b5:99:00:00:00)

> Internet Protocol Version 4, Src: 172.17.93.192, Dst: 142.251.221.164

> User Datagram Protocol, Src Port: 55892, Dst Port: 443

> QUIC IETF

0000 68 b5 99 ce 84 1a 68 ca c4 a8 84 58 08 00 45 00 h...h...X..E..

0010 04 fe 00 00 40 00 48 11 bf 7d ac 11 5d c0 8e fb ...@...}...]

0020 dd a4 de 54 01 bb 04 ea b9 5e cd 00 00 00 01 00 ...T...^...h...

0030 84 9c f2 14 5a 03 ec d2 00 00 44 00 f7 5a 48 f2 ...Z...D...ZH...

0040 03 64 21 87 90 c8 e7 64 a8 b8 27 df 94 9b 25 ee ...d!...p...f...

0050 bd 8a a1 8d 0d 18 b3 70 de 9b a7 bc e5 0b 7b 1d ...p...p...f...

0060 c8 6f cf 22 61 5c 93 1d cc 64 c2 68 9a 27 ca 2b ...o"b\...d.h'+...

0070 ee f0 fe 97 32 0b 20 66 55 b4 8e 3a 60 27 fe 70 ...-2.f.U...K'x...

0080 54 29 99 df b6 c9 83 74 37 aa 6e 15 15 1b 0d a0 ...T...t 7 n...r...

0090 c3 83 0f 92 97 1b c6 be 42 19 fa 88 6f 09 4d c8 ...B...o.M...

00a0 b3 55 7a 70 5f 2c 45 0e 5d 61 52 ed 3a 70 b8 55 ...Uzp...E..JaR...p.U

00b0 03 4f f3 7e 07 c9 35 0e 4f ee 2a da c4 35 44 2f ...0...S...0...SD/...

00c0 90 dd 05 d2 f6 31 6d 4a c8 c3 d0 1d 58 b3 cd 04 ......mJ...X...

00d0 ef 01 66 69 a1 b6 21 ac 85 20 79 5b c8 e7 83 72 ...f1.../...yI...r...

00e0 d0 b8 3e e8 b2 62 14 cf 0a 89 77 7e 7a a0 84 83 ...>...b...v.....

00f0 d1 90 21 22 cc 72 b6 ee 5a 32 14 95 ff 03 f1 93 ...!M...p.ZZ.....

0100 30 8e d0 41 60 2f d1 b9 fd 77 8b 79 44 2b ee c3 ...0...A.../...w.yD+...

0110 93 ea 30 06 2d 95 d4 c5 a1 48 25 06 3b 5f 80 94 ...0...-...H...

0120 0c 42 0e 9c 00 6d 89 18 60 ed 6c 83 ea 97 59 e9 ...B...m...^L...Y...

0130 5e 1a ff 3a 95 b1 92 8a a2 f9 91 c4 e5 b3 2b 0d ...^!...:.....+

0140 3d 90 20 80 dc e7 99 ed 5b 61 86 f4 c7 cf 1e 9d ...=...[a].....M...

0150 9b 11 71 14 f3 83 41 2d cc ca f2 e6 8c 4d cc dc ...q...A...-...M...

0160 b5 0c 96 ca d0 2c 09 48 43 67 63 e9 7b fa a5 ea ...-...H.Cgc...f...

0170 e1 ea 23 3f d1 11 62 16 a3 13 74 91 35 bf 2b a1 ...?..b...t...5+...

0180 10 5e a1 70 1f a0 e7 d4 a5 5e d3 d5 4e 39 58 11 ...^...{...^..N9X...

(using quiche) (quiche\_traffic.pcap) (quiche\_keys.log) (debug.sqlog)

| No. | Time      | Source        | Destination   | Protocol | Length | Info                                                                                                                     |
|-----|-----------|---------------|---------------|----------|--------|--------------------------------------------------------------------------------------------------------------------------|
| 9   | 7.045833  | 172.17.93.192 | 142.251.43.67 | QUIC     | 1292   | Initial, DCID=2967cf92f36594a2, PKN: 1, PADDING, PING, CRYPTO, CRYPTO, PADDING                                           |
| 7   | 7.045886  | 172.17.93.192 | 142.251.43.67 | QUIC     | 1292   | Initial, DCID=2967cf92f36594a2, PKN: 2, CRYPTO, PING, PADDING, PING, PING, CRYPTO                                        |
| 11  | 7.046166  | 172.17.93.192 | 142.251.43.67 | QUIC     | 124    | 0-RTT, DCID=2967cf92f36594a2                                                                                             |
| 12  | 7.046681  | 172.17.93.192 | 142.251.43.67 | QUIC     | 724    | 0-RTT, DCID=2967cf92f36594a2                                                                                             |
| 13  | 7.055759  | 142.251.43.67 | 172.17.93.192 | QUIC     | 82     | Initial, SCID=e967cf92f36594a2, PKN: 1, ACK                                                                              |
| 14  | 7.055763  | 142.251.43.67 | 172.17.93.192 | QUIC     | 1292   | Initial, SCID=e967cf92f36594a2, PKN: 2, ACK, PADDING                                                                     |
| 15  | 7.068945  | 172.17.93.192 | 142.251.43.67 | QUIC     | 1292   | Initial, DCID=e967cf92f36594a2, PKN: 6, PADDING, PING, PADDING                                                           |
| 16  | 7.088135  | 142.251.43.67 | 172.17.93.192 | QUIC     | 1292   | Initial, SCID=e967cf92f36594a2, PKN: 3, CRYPTO, PADDING                                                                  |
| 17  | 7.088140  | 142.251.43.67 | 172.17.93.192 | QUIC     | 320    | Protected Payload (KP0)                                                                                                  |
| 18  | 7.089660  | 172.17.93.192 | 142.251.43.67 | QUIC     | 1292   | Handshake, DCID=e967cf92f36594a2                                                                                         |
| 23  | 7.094410  | 142.251.43.67 | 172.17.93.192 | QUIC     | 1292   | Initial, SCID=e967cf92f36594a2, PKN: 9, ACK, CRYPTO, PADDING                                                             |
| 47  | 13.817202 | 172.17.93.192 | 104.18.27.14  | QUIC     | 1242   | Forcing Version Negotiation, DCID=b5e91fe4fb6186a9db8bf62774fca57, SCID=e994bc8ad7c676441f0c6a7c45239bfe7dc32f06, PKN: 1 |
| 48  | 13.825061 | 104.18.27.14  | 172.17.93.192 | QUIC     | 89     | Version Negotiation, DCID=e994bc8ad7c676441f0c6a7c45239bfe7dc32f06, SCID=b5e91fe4fb6186a9db8bf62774fca57, PKN: 1         |
| 49  | 13.825595 | 172.17.93.192 | 104.18.27.14  | QUIC     | 1242   | Initial, DCID=b5e91fe4fb6186a9db8bf62774fca57, SCID=e994bc8ad7c676441f0c6a7c45239bfe7dc32f06, PKN: 1                     |
| 50  | 13.836659 | 104.18.27.14  | 172.17.93.192 | QUIC     | 1242   | Initial, DCID=e994bc8ad7c676441f0c6a7c45239bfe7dc32f06, SCID=b16f2e97d3c0c6a7c45239bfe7dc32f06, PKN: 1                   |
| 51  | 13.863681 | 104.18.27.14  | 172.17.93.192 | QUIC     | 1242   | Initial, DCID=e994bc8ad7c676441f0c6a7c45239bfe7dc32f06, SCID=b16f2e97d3c0c6a7c45239bfe7dc32f06, PKN: 1                   |
| 52  | 13.863684 | 104.18.27.14  | 172.17.93.192 | QUIC     | 1242   | Handshake, DCID=e994bc8ad7c676441f0c6a7c45239bfe7dc32f06, SCID=b16f2e97d3c0c6a7c45239bfe7dc32f06, PKN: 1                 |
| 53  | 13.863686 | 104.18.27.14  | 172.17.93.192 | QUIC     | 630    | Handshake, DCID=e994bc8ad7c676441f0c6a7c45239bfe7dc32f06, SCID=b16f2e97d3c0c6a7c45239bfe7dc32f06, PKN: 1                 |
| 54  | 13.863688 | 104.18.27.14  | 172.17.93.192 | QUIC     | 1242   | Handshake, DCID=e994bc8ad7c676441f0c6a7c45239bfe7dc32f06, SCID=b16f2e97d3c0c6a7c45239bfe7dc32f06, PKN: 1                 |
| 55  | 13.868410 | 172.17.93.192 | 104.18.27.14  | HTTP3    | 1392   | SETTINGS, PADDING                                                                                                        |
| 56  | 13.868440 | 172.17.93.192 | 104.18.27.14  | HTTP3    | 86     |                                                                                                                          |
| 57  | 13.868449 | 172.17.93.192 | 104.18.27.14  | HTTP3    | 86     |                                                                                                                          |
| 58  | 13.895330 | 172.17.93.192 | 104.18.27.14  | HTTP3    | 152    | HEADERS: GET https://cloudflare-quic.com/                                                                                |
| 59  | 13.877084 | 104.18.27.14  | 172.17.93.192 | HTTP3    | 86     |                                                                                                                          |
| 60  | 13.877886 | 104.18.27.14  | 172.17.93.192 | HTTP3    | 595    | SETTINGS                                                                                                                 |
| 62  | 13.878892 | 104.18.27.14  | 172.17.93.192 | HTTP3    | 92     |                                                                                                                          |
| 63  | 13.878894 | 104.18.27.14  | 172.17.93.192 | HTTP3    | 111    |                                                                                                                          |
| 65  | 13.888810 | 104.18.27.14  | 172.17.93.192 | HTTP3    | 246    | HEADERS: 200 OK                                                                                                          |
| 66  | 13.888812 | 104.18.27.14  | 172.17.93.192 | QUIC     | 1242   | Protected Payload (KP0), DCID=e994bc8ad7c676441f0c6a7c45239bfe7dc32f06, PKN: 1                                           |
| 100 | 13.910297 | 104.18.27.14  | 172.17.93.192 | HTTP3    | 1242   | Unknown (0x2f72)                                                                                                         |

> Frame 58: Packet, 152 bytes on wire (1216 bits), 152 bytes captured (1216 bits) on interface 0

> Ethernet II, Src: Apple\_a8:84:58 (68:ca:c4:a8:84:58), Dst: Hewlett-Packard\_ce:B4:1a (68:b5:99:00:00:00)

> Internet Protocol Version 4, Src: 172.17.93.192, Dst: 104.18.27.14

> User Datagram Protocol, Src Port: 51741, Dst Port: 443

> QUIC IETF

> QUIC Connection Information

> [Packet Length: 118]

> QUIC Short Header: DCID=b16f2e97d3c0c6a7c45239bfe7dc32f06, PKN=7

> STREAM Id=0 fin=1 off=0 len=67 dir=bidirectional origin=client-initiated

> Hypertext Transfer Protocol Version 3

0000 68 b5 99 ce 84 1a 68 ca c4 a8 84 58 08 00 45 00 h...h...X..E..

0010 00 8a fe 00 00 40 00 48 11 ee ac 11 5d c0 68 12 ...-...v...E...e...

0020 1b 0e ca 1d 01 bb 00 76 d6 dc 45 01 6f 2e 97 d3 ...-...v...X...^...

0030 c0 c4 aa 56 6d 03 97 99 c0 c6 58 ae a2 25 c1 5e ...-...V...-...P...

0040 ed 97 d3 3e a9 e4 a4 91 72 49 79 88 50 9c a1 a3 ...>...-...rIY.P...

0050 2b ff 7a 98 51 1d 62 a4 80 32 d1 10 6b b4 31 ac ...+z.Q.b...2..k.1...

0060 f3 1d a7 71 12 2e c0 5c bf e1 22 f0 a5 39 36 31 ...-...q...^...m..961...

0070 99 ca 33 fb de 3f f7 62 44 05 c0 f1 94 81 27 c5 ...-...S...7..b...D...-

0080 3a 6d 2b e9 66 18 7c 18 9b 81 4e 5d 09 9d 95 13 ...+...f...J...N].....

0090 9c ae c7 f5 88 9b a4 45 .....

## SQLLOG File:

```
← → logs
debug.sqllog x
debug.sqllog
1 [{"qlog_version":"0.3","qlog_format":"JSON-SEQ","title":"quiche-client qlog","description":"quiche-client qlog id=c152e01e38a57
2 [{"time":0.0,"name":"transport:parameters_set","data":{"owner":"local","disable_active_migration":true,"max_idle_timeout":30000
3 [{"time":0.962334,"name":"transport:packet_sent","data":{"header":{"packet_type":"initial","packet_number":0,"version":"bababat
4 [{"time":0.962334,"name":"recovery:metrics_updated","data":{"cf_lost_bytes":{"total":0,"delta":0},"cf_lost_packets":{"total":0,
5 [{"time":0.962334,"name":"recovery:congestion_state_updated","data":{"new":"slow_start"}}
6 [{"time":18.211,"name":"transport:packet_sent","data":{"header":{"packet_type":"initial","packet_number":1,"version":"1","scil
7 [{"time":61.422546,"name":"transport:packet_received","data":{"header":{"packet_type":"initial","packet_number":0,"version":"1'
8 [{"time":61.422546,"name":"recovery:metrics_updated","data":{"min_rtt":43.211544,"smoothed_rtt":43.211544,"latest_rtt":43.21154
9 [{"time":62.685123,"name":"transport:packet_received","data":{"header":{"packet_type":"handshake","packet_number":1,"version":'
10 [{"time":62.766163,"name":"transport:packet_received","data":{"header":{"packet_type":"handshake","packet_number":2,"version":'
11 [{"time":62.806583,"name":"transport:packet_received","data":{"header":{"packet_type":"handshake","packet_number":3,"version":'
12 [{"time":62.806583,"name":"transport:parameters_set","data":{"owner":"remote","tls_cipher":"AES128_GCM","original_destination_c
13 [{"time":66.238335,"name":"transport:data_moved","data":{"stream_id":2,"offset":0,"length":1,"from":"application","to":"transp
14 [{"time":66.24946,"name":"http:stream_type_set","data":{"owner":"local","stream_id":2,"stream_type":"control"}}
15 [{"time":66.41367,"name":"transport:data_moved","data":{"stream_id":2,"offset":1,"length":18,"from":"application","to":"transp
16 [{"time":66.42379,"name":"http:frame_created","data":{"stream_id":2,"length":18,"frame":{"frame_type":"settings","settings":[{"
17 [{"time":66.4417,"name":"transport:data_moved","data":{"stream_id":6,"offset":0,"length":1,"from":"application","to":"transport
18 [{"time":66.44908,"name":"http:stream_type_set","data":{"owner":"local","stream_id":6,"stream_type":"qpack_encode"}}
19 [{"time":66.45929,"name":"transport:data_moved","data":{"stream_id":10,"offset":0,"length":1,"from":"application","to":"transp
20 [{"time":66.46662,"name":"http:stream_type_set","data":{"owner":"local","stream_id":10,"stream_type":"qpack_decode"}}
21 [{"time":66.625206,"name":"transport:data_moved","data":{"stream_id":0,"offset":0,"length":0,"from":"application","to":"transp
22 [{"time":66.64137,"name":"transport:data_moved","data":{"stream_id":0,"offset":0,"length":8,"from":"application","to":"transport
23 [{"time":66.650665,"name":"transport:data_moved","data":{"stream_id":0,"offset":8,"length":1,"from":"application","to":"transport
24 [{"time":66.65817,"name":"http:frame_created","data":{"stream_id":0,"length":0,"frame":{"frame_type":"reserved","length":0}}}
25 [{"time":66.66621,"name":"transport:data_moved","data":{"stream_id":0,"offset":9,"length":8,"from":"application","to":"transport
26 [{"time":66.67425,"name":"transport:data_moved","data":{"stream_id":0,"offset":17,"length":1,"from":"application","to":"transport
27 [{"time":66.68259,"name":"transport:data_moved","data":{"stream_id":0,"offset":18,"length":18,"from":"application","to":"transport
28 [{"time":66.690125,"name":"http:frame_created","data":{"stream_id":0,"length":18,"frame":{"frame_type":"reserved","length":18}}
29 [{"time":66.708786,"name":"transport:data_moved","data":{"stream_id":0,"offset":36,"length":2,"from":"application","to":"transport
30 [{"time":66.71762,"name":"transport:data_moved","data":{"stream_id":0,"offset":38,"length":29,"from":"application","to":"transport
31 [{"time":66.72771,"name":"http:frame_created","data":{"stream_id":0,"length":29,"frame":{"frame_type":"headers","headers":[{"na
32 [{"time":66.77859,"name":"transport:packet_sent","data":{"header":{"packet_type":"initial","packet_number":2,"version":"1","sci
33 [{"time":66.77859,"name":"transport:packet_sent","data":{"header":{"packet_type":"handshake","packet_number":3,"version":"1","s
34 [{"time":66.77859,"name":"recovery:metrics_updated","data":{"bytes_in_flight":112}}
35 [{"time":66.77859,"name":"transport:packet_sent","data":{"header":{"packet_type":"1RTT","packet_number":4,"raw":{"length":1165
36 [{"time":66.77859,"name":"recovery:metrics_updated","data":{"bytes_in_flight":1277}}
37 [{"time":67.114075,"name":"transport:packet_sent","data":{"header":{"packet_type":"1RTT","packet_number":5,"raw":{"length":44,
38 [{"time":67.114075,"name":"recovery:metrics_updated","data":{"bytes_in_flight":1321}}
39 [{"time":67.16813,"name":"transport:packet_sent","data":{"header":{"packet_type":"1RTT","packet_number":6,"raw":{"length":44,"
40 [{"time":67.16813,"name":"recovery:metrics_updated","data":{"bytes_in_flight":1365}}
41 [{"time":67.21338,"name":"transport:packet_sent","data":{"header":{"packet_type":"1RTT","packet_number":7,"raw":{"length":110,
42 [{"time":67.21338,"name":"recovery:metrics_updated","data":{"bytes_in_flight":1475}}
43 [{"time":78.97446,"name":"transport:packet_received","data":{"header":{"packet_type":"1RTT","packet_number":4,"raw":{"length":
44 [{"time":78.97446,"name":"recovery:metrics_updated","data":{"min_rtt":11.761085,"smoothed_rtt":39.28023,"latest_rtt":11.761085,
```

## (using ngtcp2) (ngtcp2\_traffic.pcap) (ngtcp2\_keys.log) (debug.sqllog)

|     |           |                 |                 |        |      |                                                                                         |
|-----|-----------|-----------------|-----------------|--------|------|-----------------------------------------------------------------------------------------|
| 60  | 6.491241  | 104.18.26.14    | 172.17.93.192   | QUIC   | 1294 | Initial, DCID=db89df631b9e32c839884af3d7d844b27a, SCID=011f153b7bedf2fa21de83b6ee...    |
| 69  | 6.491241  | 104.18.26.14    | 172.17.93.192   | QUIC   | 621  | Handshake, DCID=db89df631b9e32c839884af3d7d844b27a, SCID=011f153b7bedf2fa21de83b6ee...  |
| 70  | 6.500890  | 172.17.93.192   | 104.18.26.14    | HTTP/3 | 261  | SETTINGS, STREAM(10), STREAM(6)                                                         |
| 74  | 7.805107  | 104.18.26.14    | 172.17.93.192   | QUIC   | 1242 | Initial, DCID=db89df631b9e32c839884af3d7d844b27a, SCID=011f153b7bedf2fa21de83b6ee...    |
| 75  | 7.805112  | 104.18.26.14    | 172.17.93.192   | QUIC   | 1242 | Initial, DCID=db89df631b9e32c839884af3d7d844b27a, SCID=011f153b7bedf2fa21de83b6ee...    |
| 76  | 7.972632  | 172.17.93.192   | 104.18.26.14    | QUIC   | 149  | Handshake, DCID=011f153b7bedf2fa21de83b6eedf8d26a06ea9f, SCID=db89df631b9e32c839884...  |
| 77  | 7.973486  | 172.17.93.192   | 104.18.26.14    | QUIC   | 149  | Handshake, DCID=011f153b7bedf2fa21de83b6eedf8d26a06ea9f, SCID=db89df631b9e32c839884...  |
| 81  | 9.883445  | 104.18.26.14    | 172.17.93.192   | QUIC   | 1242 | Handshake, DCID=db89df631b9e32c839884af3d7d844b27a, SCID=011f153b7bedf2fa21de83b6ee...  |
| 82  | 9.883449  | 104.18.26.14    | 172.17.93.192   | QUIC   | 1242 | Handshake, DCID=db89df631b9e32c839884af3d7d844b27a, SCID=011f153b7bedf2fa21de83b6ee...  |
| 83  | 9.884135  | 172.17.93.192   | 104.18.26.14    | QUIC   | 117  | Handshake, DCID=011f153b7bedf2fa21de83b6eedf8d26a06ea9f, SCID=db89df631b9e32c839884...  |
| 86  | 10.913889 | 172.17.93.192   | 104.18.26.14    | QUIC   | 149  | Handshake, DCID=011f153b7bedf2fa21de83b6eedf8d26a06ea9f, SCID=db89df631b9e32c839884...  |
| 87  | 10.914628 | 172.17.93.192   | 104.18.26.14    | QUIC   | 149  | Handshake, DCID=011f153b7bedf2fa21de83b6eedf8d26a06ea9f, SCID=db89df631b9e32c839884...  |
| 108 | 16.788201 | 172.17.93.192   | 104.18.26.14    | QUIC   | 149  | Handshake, DCID=011f153b7bedf2fa21de83b6eedf8d26a06ea9f, SCID=db89df631b9e32c839884...  |
| 109 | 16.788899 | 172.17.93.192   | 104.18.26.14    | QUIC   | 149  | Handshake, DCID=011f153b7bedf2fa21de83b6eedf8d26a06ea9f, SCID=db89df631b9e32c839884...  |
| 114 | 17.839323 | 104.18.26.14    | 172.17.93.192   | HTTP/3 | 599  | SETTINGS                                                                                |
| 115 | 17.839327 | 104.18.26.14    | 172.17.93.192   | HTTP/3 | 83   |                                                                                         |
| 116 | 17.839329 | 104.18.26.14    | 172.17.93.192   | HTTP/3 | 83   |                                                                                         |
| 117 | 17.844858 | 172.17.93.192   | 104.18.26.14    | HTTP/3 | 151  | SETTINGS, STREAM(6), STREAM(10)                                                         |
| 118 | 17.860758 | 104.18.26.14    | 172.17.93.192   | HTTP/3 | 108  |                                                                                         |
| 168 | 80.571559 | 172.17.93.192   | 142.251.222.202 | QUIC   | 1292 | Initial, DCID=912e7d20ae41d3ad, PKN: 1, CRYPTO, PADDING, CRYPTO, CRYPTO, PADDING, PI... |
| 169 | 80.571560 | 172.17.93.192   | 142.251.222.202 | QUIC   | 1292 | Initial, DCID=912e7d20ae41d3ad, PKN: 2, CRYPTO                                          |
| 170 | 80.571592 | 172.17.93.192   | 142.251.222.202 | QUIC   | 1292 | Initial, DCID=912e7d20ae41d3ad, PKN: 3, CRYPTO, PADDING, CRYPTO, PADDING, CRYPTO, PA... |
| 171 | 80.578729 | 172.17.93.192   | 142.251.222.202 | QUIC   | 121  | 0-RTT, DCID=912e7d20ae41d3ad, PKN: 2, ACK                                               |
| 172 | 80.585839 | 142.251.222.202 | 172.17.93.192   | QUIC   | 82   | Initial, SCID=f12e7d20ae41d3ad, PKN: 1, ACK                                             |
| 173 | 80.585842 | 142.251.222.202 | 172.17.93.192   | QUIC   | 82   | Initial, SCID=f12e7d20ae41d3ad, PKN: 2, ACK                                             |
| 174 | 80.587835 | 142.251.222.202 | 172.17.93.192   | QUIC   | 1292 | Initial, SCID=f12e7d20ae41d3ad, PKN: 3, ACK, PADDING                                    |
| 175 | 80.587837 | 142.251.222.202 | 172.17.93.192   | QUIC   | 1292 | Initial, SCID=f12e7d20ae41d3ad, PKN: 4, ACK, PADDING                                    |
| 177 | 80.596931 | 172.17.93.192   | 142.251.222.202 | QUIC   | 1292 | Initial, DCID=f12e7d20ae41d3ad, PKN: 6, PADDING, PING, PADDING                          |
| 178 | 80.617642 | 142.251.222.202 | 172.17.93.192   | QUIC   | 1292 | Initial, SCID=f12e7d20ae41d3ad, PKN: 5, CRYPTO, PADDING                                 |
| 179 | 80.617645 | 142.251.222.202 | 172.17.93.192   | QUIC   | 315  | Protected Payload (KP0)                                                                 |

> Frame 2: Packet, 1292 bytes on wire (10336 bits), 1292 bytes captured (10336 bits) on interface 0

> Ethernet II, Src: Apple\_a8:84:58 (68:ca:c4:a8:84:58), Dst: HewlettPacka\_ce:84:1a (68:b5:14:ce:84:1a)

> Internet Protocol Version 4, Src: 172.17.93.192, Dst: 216.239.38.223

> User Datagram Protocol, Src Port: 63388, Dst Port: 443

▼ QUIC IETF

> QUIC Connection Information

[Packet Length: 1292]

1... .. = Header Form: Long Header (1)

..1... .. = Fixed Bits: True

..00... .. = Packet Type: Initial (0)

[... ..00... .. = Reserved: 0]

[... ..00... .. = Packet Number Length: 1 bytes (0)]

0080 ae 44 89 a5 36 62 75 af fd ec 72 19 69 ea 6f f8 ..D..6bu...r...o

0081 03 dd bf 43 0b 57 6e 8c e4 e6 5e 56 87 1e 5f ba ...C.Wm...v...-

00a0 07 49 3a 1d 15 36 57 18 ee 02 7d 5c b8 4c d6 ab ..I..6W...}\..L..

00b0 72 cd 4f 58 5a e1 ab ad 60 41 8e ce 6e a1 cb 7c ..r.OXZ...}\..n..

00c0 cc 69 1b be 71 e8 19 1f 01 57 3e b7 61 4e 76 fd ..i..q...W..aNV..

00d0 c3 a3 91 c8 5c 00 c0 f6 21 b6 d2 a2 4f c0 bb b1 .....\...i...0...

00e0 0f 42 65 04 51 cb b4 9c ae 25 e2 07 16 20 88 2f ..Be.Q...A...(\./

00f0 ce 04 3a 56 83 ee c0 19 91 c2 a3 97 0f 4e 44 da ...V...ND...

0100 b5 f4 b8 42 f5 8d 06 28 79 dd a9 7e 03 d0 90 8f ...B...(\...y...-

0110 61 0f 75 8e b7 6e 57 b1 e9 19 2b a1 c4 da 75 79 ..a..nW...uy

0120 b6 8e f3 2f 31 c3 6e 31 35 2d cf 06 3d 8d 87 43 .../..1..5...-c..

0130 6f 0b 07 61 48 14 f3 9b ee 03 97 0b fc 01 88 b2 ..o..aH...-.....

0140 ca 4d 15 ae ea 2d 08 59 d5 e0 ab 8d 30 f3 03 0c ..N...-Y...T...-

0150 fe d6 7d f2 4e d5 6b 5d 8f 54 05 0d 09 a1 17 9e ...}..N..k}...T...-

0160 65 a7 74 ee 38 f4 27 e8 46 2e 95 cc 7c 1a f4 b7 ...t..B...L...F...-

## SQLLOG File:

|              |                                                                                                                                |        |   |
|--------------|--------------------------------------------------------------------------------------------------------------------------------|--------|---|
| ← →          |                                                                                                                                | Q logs | ↻ |
| debug.sqllog |                                                                                                                                |        |   |
| debug.sqllog |                                                                                                                                |        |   |
| 1            | ["qlog_format":"JSON-SEQ","qlog_version":"0.3","trace":{"vantage_point":{"name":"ngtcp2","type":"client"},"common_fields":{"pr |        |   |
| 2            | ["time":0,"name":"transport:parameters_set","data":{"owner":"local","initial_source_connection_id":"1078e2cc0cd6e2b1b2aff73b25 |        |   |
| 3            | ["time":1,"name":"transport:packet_sent","data":{"frames":[{"frame_type":"crypto","offset":94,"length":4},{"frame_type":"crypt |        |   |
| 4            | ["time":1,"name":"recovery:metrics_updated","data":{"smoothed_rtt":333,"latest_rtt":0,"rtt_variance":166,"pto_count":0,"conges |        |   |
| 5            | ["time":21,"name":"transport:packet_received","data":{"frames":[{"frame_type":"ack","ack_delay":0,"acked_ranges":[{"1779950144 |        |   |
| 6            | ["time":21,"name":"recovery:metrics_updated","data":{"min_rtt":20,"smoothed_rtt":20,"latest_rtt":20,"rtt_variance":10,"pto_cou |        |   |
| 7            | ["time":21,"name":"transport:packet_received","data":{"frames":[{"frame_type":"crypto","offset":0,"length":56},"header":{"pad  |        |   |
| 8            | ["time":21,"name":"recovery:metrics_updated","data":{"min_rtt":20,"smoothed_rtt":20,"latest_rtt":20,"rtt_variance":10,"pto_cou |        |   |
| 9            | ["time":22,"name":"transport:packet_sent","data":{"frames":[{"frame_type":"ack","ack_delay":0,"acked_ranges":[{"0,1}],{"frame  |        |   |
| 10           | ["time":22,"name":"recovery:metrics_updated","data":{"min_rtt":20,"smoothed_rtt":20,"latest_rtt":20,"rtt_variance":10,"pto_cou |        |   |
| 11           | ["time":22,"name":"transport:packet_sent","data":{"frames":[{"frame_type":"crypto","offset":442,"length":1},{"frame_type":"cry |        |   |
| 12           | ["time":22,"name":"recovery:metrics_updated","data":{"min_rtt":20,"smoothed_rtt":20,"latest_rtt":20,"rtt_variance":10,"pto_cou |        |   |
| 13           | ["time":30,"name":"transport:packet_received","data":{"frames":[{"frame_type":"ack","ack_delay":0,"acked_ranges":[{"1779950145 |        |   |
| 14           | ["time":30,"name":"recovery:metrics_updated","data":{"min_rtt":8,"smoothed_rtt":19,"latest_rtt":8,"rtt_variance":10,"pto_cou   |        |   |
| 15           | ["time":31,"name":"transport:packet_received","data":{"frames":[{"frame_type":"ack","ack_delay":0,"acked_ranges":[{"1779950145 |        |   |
| 16           | ["time":31,"name":"recovery:metrics_updated","data":{"min_rtt":8,"smoothed_rtt":18,"latest_rtt":8,"rtt_variance":10,"pto_cou   |        |   |
| 17           | ["time":31,"name":"transport:packet_received","data":{"frames":[{"frame_type":"crypto","offset":1176,"length":58},"header":{"  |        |   |
| 18           | ["time":31,"name":"recovery:metrics_updated","data":{"min_rtt":8,"smoothed_rtt":18,"latest_rtt":8,"rtt_variance":10,"pto_cou   |        |   |
| 19           | ["time":31,"name":"transport:packet_sent","data":{"frames":[{"frame_type":"ack","ack_delay":0,"acked_ranges":[{"12,4}],{"frame |        |   |
| 20           | ["time":31,"name":"recovery:metrics_updated","data":{"min_rtt":8,"smoothed_rtt":18,"latest_rtt":8,"rtt_variance":10,"pto_cou   |        |   |
| 21           | ["time":32,"name":"transport:packet_received","data":{"frames":[{"frame_type":"crypto","offset":2265,"length":511},"header":{" |        |   |
| 22           | ["time":32,"name":"recovery:metrics_updated","data":{"min_rtt":8,"smoothed_rtt":18,"latest_rtt":8,"rtt_variance":10,"pto_cou   |        |   |
| 23           | ["time":32,"name":"transport:packet_sent","data":{"frames":[{"frame_type":"ack","ack_delay":0,"acked_ranges":[{"7}],{"header": |        |   |
| 24           | ["time":32,"name":"recovery:metrics_updated","data":{"min_rtt":8,"smoothed_rtt":18,"latest_rtt":8,"rtt_variance":10,"pto_cou   |        |   |
| 25           | ["time":32,"name":"transport:packet_received","data":{"frames":[{"frame_type":"crypto","offset":1133,"length":1132},"header":  |        |   |
| 26           | ["time":32,"name":"recovery:metrics_updated","data":{"min_rtt":8,"smoothed_rtt":18,"latest_rtt":8,"rtt_variance":10,"pto_cou   |        |   |
| 27           | ["time":32,"name":"transport:packet_sent","data":{"frames":[{"frame_type":"ack","ack_delay":0,"acked_ranges":[{"6,7}],{"header |        |   |
| 28           | ["time":32,"name":"recovery:metrics_updated","data":{"min_rtt":8,"smoothed_rtt":18,"latest_rtt":8,"rtt_variance":10,"pto_cou   |        |   |
| 29           | ["time":34,"name":"transport:parameters_set","data":{"owner":"remote","initial_source_connection_id":"01607d96cd7c77ef4604696  |        |   |
| 30           | ["time":34,"name":"transport:packet_received","data":{"frames":[{"frame_type":"crypto","offset":0,"length":1133},"header":{"   |        |   |
| 31           | ["time":34,"name":"recovery:metrics_updated","data":{"min_rtt":8,"smoothed_rtt":18,"latest_rtt":8,"rtt_variance":10,"pto_cou   |        |   |
| 32           | ["time":37,"name":"transport:packet_sent","data":{"frames":[{"frame_type":"ack","ack_delay":0,"acked_ranges":[{"5,7}],{"frame  |        |   |
| 33           | ["time":37,"name":"recovery:metrics_updated","data":{"min_rtt":8,"smoothed_rtt":18,"latest_rtt":8,"rtt_variance":10,"pto_cou   |        |   |
| 34           | ["time":37,"name":"transport:packet_sent","data":{"frames":[{"frame_type":"new_connection_id","sequence_number":1,"retire_prid |        |   |
| 35           | ["time":37,"name":"recovery:metrics_updated","data":{"min_rtt":8,"smoothed_rtt":18,"latest_rtt":8,"rtt_variance":10,"pto_cou   |        |   |
| 36           | ["time":37,"name":"transport:packet_sent","data":{"frames":[{"frame_type":"ping"},{"frame_type":"padding"},"header":{"packet   |        |   |
| 37           | ["time":37,"name":"recovery:metrics_updated","data":{"min_rtt":8,"smoothed_rtt":18,"latest_rtt":8,"rtt_variance":10,"pto_cou   |        |   |
| 38           | ["time":43,"name":"transport:packet_received","data":{"frames":[{"frame_type":"stream","stream_id":7,"offset":0,"length":1}],  |        |   |
| 39           | ["time":43,"name":"recovery:metrics_updated","data":{"min_rtt":8,"smoothed_rtt":18,"latest_rtt":8,"rtt_variance":10,"pto_cou   |        |   |
| 40           | ["time":43,"name":"transport:packet_received","data":{"frames":[{"frame_type":"ack","ack_delay":0,"acked_ranges":[{"1779950144 |        |   |
| 41           | ["time":43,"name":"recovery:metrics_updated","data":{"min_rtt":6,"smoothed_rtt":16,"latest_rtt":6,"rtt_variance":10,"pto_cou   |        |   |
| 42           | ["time":43,"name":"transport:packet_received","data":{"frames":[{"frame_type":"stream","stream_id":11,"offset":0,"length":1}], |        |   |
| 43           | ["time":43,"name":"recovery:metrics_updated","data":{"min_rtt":6,"smoothed_rtt":16,"latest_rtt":6,"rtt_variance":10,"pto_cou   |        |   |
| 44           | ["time":43,"name":"transport:packet_received","data":{"frames":[{"frame_type":"stream","stream_id":15,"offset":0,"length":26,1 |        |   |



# NON-QUIC DATA

## DNS

| No. | Time     | Source               | Destination                           | Protocol | Length | Info                                                                          |
|-----|----------|----------------------|---------------------------------------|----------|--------|-------------------------------------------------------------------------------|
| 1   | 0.000000 | 2409:40f4:a7:3f71::  | 64:ff9b::808:808                      | DNS      | 101    | Standard query 0x3775 A google.com OPT                                        |
| 2   | 0.043857 | 64:ff9b::808:808     | 2409:40f4:a7:3f71::446:617c:2862:27b7 | DNS      | 117    | Standard query response 0x3775 A google.com A 142.250.67.46 OPT               |
| 3   | 0.048828 | 2409:40f4:a7:3f71::  | 64:ff9b::808:808                      | DNS      | 101    | Standard query 0x10e6 A google.com OPT                                        |
| 4   | 0.116159 | 64:ff9b::808:808     | 2409:40f4:a7:3f71::446:617c:2862:27b7 | DNS      | 117    | Standard query response 0x10e6 A google.com A 142.250.67.46 OPT               |
| 5   | 0.116412 | 2409:40f4:a7:3f71::  | 64:ff9b::808:808                      | DNS      | 101    | Standard query 0xb6b15 A google.com OPT                                       |
| 6   | 0.154997 | 64:ff9b::808:808     | 2409:40f4:a7:3f71::446:617c:2862:27b7 | DNS      | 117    | Standard query response 0xb6b15 A google.com A 172.217.24.142 OPT             |
| 7   | 0.161895 | 2409:40f4:a7:3f71::  | 64:ff9b::808:808                      | DNS      | 101    | Standard query 0x4b27 A google.com OPT                                        |
| 8   | 0.199987 | 64:ff9b::808:808     | 2409:40f4:a7:3f71::446:617c:2862:27b7 | DNS      | 117    | Standard query response 0x4b27 A google.com A 142.251.222.142 OPT             |
| 9   | 0.208329 | 2409:40f4:a7:3f71::  | 64:ff9b::808:808                      | DNS      | 101    | Standard query 0xb36 A google.com OPT                                         |
| 10  | 0.279089 | 64:ff9b::808:808     | 2409:40f4:a7:3f71::446:617c:2862:27b7 | DNS      | 117    | Standard query response 0xb36 A google.com A 142.250.67.46 OPT                |
| 11  | 0.281465 | 2409:40f4:a7:3f71::  | 64:ff9b::808:808                      | DNS      | 101    | Standard query 0xa5af A google.com OPT                                        |
| 12  | 0.374828 | 64:ff9b::808:808     | 2409:40f4:a7:3f71::446:617c:2862:27b7 | DNS      | 117    | Standard query response 0xa5af A google.com A 172.217.24.14 OPT               |
| 13  | 0.379743 | 2409:40f4:a7:3f71::  | 64:ff9b::808:808                      | DNS      | 101    | Standard query 0xa6a1 A google.com OPT                                        |
| 14  | 0.416503 | 64:ff9b::808:808     | 2409:40f4:a7:3f71::446:617c:2862:27b7 | DNS      | 117    | Standard query response 0xa6a1 A google.com A 142.250.67.46 OPT               |
| 15  | 0.421544 | 2409:40f4:a7:3f71::  | 64:ff9b::808:808                      | DNS      | 101    | Standard query 0x7909 A google.com OPT                                        |
| 16  | 0.493937 | 64:ff9b::808:808     | 2409:40f4:a7:3f71::446:617c:2862:27b7 | DNS      | 117    | Standard query response 0x7909 A google.com A 142.251.222.142 OPT             |
| 17  | 0.501444 | 2409:40f4:a7:3f71::  | 64:ff9b::808:808                      | DNS      | 101    | Standard query 0xd15f A google.com OPT                                        |
| 18  | 0.578184 | 64:ff9b::808:808     | 2409:40f4:a7:3f71::446:617c:2862:27b7 | DNS      | 117    | Standard query response 0xd15f A google.com A 142.250.67.46 OPT               |
| 19  | 0.591165 | 2409:40f4:a7:3f71::  | 64:ff9b::808:808                      | DNS      | 101    | Standard query 0x9165 A google.com OPT                                        |
| 20  | 0.668690 | 64:ff9b::808:808     | 2409:40f4:a7:3f71::446:617c:2862:27b7 | DNS      | 117    | Standard query response 0x9165 A google.com A 142.251.223.14 OPT              |
| 21  | 0.683388 | 2409:40f4:a7:3f71::  | 64:ff9b::808:808                      | DNS      | 101    | Standard query 0x3846 A google.com OPT                                        |
| 22  | 0.705343 | 64:ff9b::808:808     | 2409:40f4:a7:3f71::446:617c:2862:27b7 | DNS      | 117    | Standard query response 0x3846 A google.com A 142.251.222.142 OPT             |
| 23  | 0.717176 | 2409:40f4:a7:3f71::  | 64:ff9b::808:808                      | DNS      | 101    | Standard query 0x22d9 A google.com OPT                                        |
| 24  | 0.765721 | 64:ff9b::808:808     | 2409:40f4:a7:3f71::446:617c:2862:27b7 | DNS      | 117    | Standard query response 0x22d9 A google.com A 142.251.222.142 OPT             |
| 25  | 0.775524 | 2409:40f4:a7:3f71::  | 64:ff9b::808:808                      | DNS      | 101    | Standard query 0x103e A google.com OPT                                        |
| 26  | 0.815683 | 64:ff9b::808:808     | 2409:40f4:a7:3f71::446:617c:2862:27b7 | DNS      | 117    | Standard query response 0x103e A google.com A 142.251.222.142 OPT             |
| 27  | 5.926727 | fe80::186f:f4f2:58:: | fe80::bc9f:58ff:fe3e:dd64             | DNS      | 112    | Standard query 0xaf3e A mobile.events.data.microsoft.com                      |
| 28  | 5.927287 | fe80::186f:f4f2:58:: | fe80::bc9f:58ff:fe3e:dd64             | DNS      | 112    | Standard query 0xaf653 AAAA mobile.events.data.microsoft.com                  |
| 29  | 6.005471 | fe80::bc9f:58ff:fe:: | fe80::186f:f4f2:5888:3770             | DNS      | 244    | Standard query response 0xaf653 AAAA mobile.events.data.microsoft.com CNAME m |
| 30  | 6.006996 | fe80::bc9f:58ff:fe:: | fe80::186f:f4f2:5888:3770             | DNS      | 232    | Standard query response 0xaf3e A mobile.events.data.microsoft.com CNAME mobi  |

> Frame 1: Packet, 101 bytes on wire (808 bits), 101 bytes captured (808 bits) on interface

> Ethernet II, Src: d2:06:37:69:89:0b (d2:06:37:69:89:0b), Dst: bc:9f:58:3e:dd:64 (bc:9f:58:3e:dd:64)

> Internet Protocol Version 6, Src: 2409:40f4:a7:3f71::446:617c:2862:27b7, Dst: 64:ff9b::808:808

> User Datagram Protocol, Src Port: 62803, Dst Port: 53

> Domain Name System (query)

0000

bc 9f 58 3e dd 64 d2 06 37 69 89 0b 86 dd 60 00

0010

00 00 00 2f 11 40 24 09 40 f4 00 a7 3f 71 04 46

0020

61 7c 28 62 27 b7 00 64 ff 9b 00 00 00 00 00

0030

00 00 08 08 08 08 f5 53 00 35 00 2f 2a a7 37 75

0040

01 20 00 01 00 00 00 00 01 06 6f 6f 6f 6f 6c

0050

65 a3 63 5f 15 c1 94 1d d0 be 81 0e 95 76 2d b2

0060

00 00 00 00 00 00 01 00 01 00 00 29 10 00 00

## Randome Noise

| No. | Time     | Source    | Destination | Protocol | Length | Info                  |
|-----|----------|-----------|-------------|----------|--------|-----------------------|
| 1   | 0.000000 | 127.0.0.1 | 127.0.0.1   | UDP      | 1232   | 63115 → 9999 Len=1200 |
| 2   | 0.012590 | 127.0.0.1 | 127.0.0.1   | UDP      | 1232   | 63115 → 9999 Len=1200 |
| 3   | 0.025118 | 127.0.0.1 | 127.0.0.1   | UDP      | 1232   | 63115 → 9999 Len=1200 |
| 4   | 0.037645 | 127.0.0.1 | 127.0.0.1   | UDP      | 1232   | 63115 → 9999 Len=1200 |
| 5   | 0.050222 | 127.0.0.1 | 127.0.0.1   | UDP      | 1232   | 63115 → 9999 Len=1200 |
| 6   | 0.062807 | 127.0.0.1 | 127.0.0.1   | UDP      | 1232   | 63115 → 9999 Len=1200 |
| 7   | 0.075389 | 127.0.0.1 | 127.0.0.1   | UDP      | 1232   | 63115 → 9999 Len=1200 |
| 8   | 0.087914 | 127.0.0.1 | 127.0.0.1   | UDP      | 1232   | 63115 → 9999 Len=1200 |
| 9   | 0.100463 | 127.0.0.1 | 127.0.0.1   | UDP      | 1232   | 63115 → 9999 Len=1200 |
| 10  | 0.113042 | 127.0.0.1 | 127.0.0.1   | UDP      | 1232   | 63115 → 9999 Len=1200 |
| 11  | 0.125630 | 127.0.0.1 | 127.0.0.1   | UDP      | 1232   | 63115 → 9999 Len=1200 |
| 12  | 0.138195 | 127.0.0.1 | 127.0.0.1   | UDP      | 1232   | 63115 → 9999 Len=1200 |
| 13  | 0.150773 | 127.0.0.1 | 127.0.0.1   | UDP      | 1232   | 63115 → 9999 Len=1200 |
| 14  | 0.163346 | 127.0.0.1 | 127.0.0.1   | UDP      | 1232   | 63115 → 9999 Len=1200 |
| 15  | 0.175918 | 127.0.0.1 | 127.0.0.1   | UDP      | 1232   | 63115 → 9999 Len=1200 |
| 16  | 0.188693 | 127.0.0.1 | 127.0.0.1   | UDP      | 1232   | 63115 → 9999 Len=1200 |
| 17  | 0.201246 | 127.0.0.1 | 127.0.0.1   | UDP      | 1232   | 63115 → 9999 Len=1200 |
| 18  | 0.212015 | 127.0.0.1 | 127.0.0.1   | UDP      | 1232   | 63115 → 9999 Len=1200 |
| 19  | 0.224695 | 127.0.0.1 | 127.0.0.1   | UDP      | 1232   | 63115 → 9999 Len=1200 |
| 20  | 0.236780 | 127.0.0.1 | 127.0.0.1   | UDP      | 1232   | 63115 → 9999 Len=1200 |
| 21  | 0.249420 | 127.0.0.1 | 127.0.0.1   | UDP      | 1232   | 63115 → 9999 Len=1200 |
| 22  | 0.263008 | 127.0.0.1 | 127.0.0.1   | UDP      | 1232   | 63115 → 9999 Len=1200 |
| 23  | 0.275680 | 127.0.0.1 | 127.0.0.1   | UDP      | 1232   | 63115 → 9999 Len=1200 |
| 24  | 0.287588 | 127.0.0.1 | 127.0.0.1   | UDP      | 1232   | 63115 → 9999 Len=1200 |
| 25  | 0.298337 | 127.0.0.1 | 127.0.0.1   | UDP      | 1232   | 63115 → 9999 Len=1200 |
| 26  | 0.310922 | 127.0.0.1 | 127.0.0.1   | UDP      | 1232   | 63115 → 9999 Len=1200 |
| 27  | 0.323478 | 127.0.0.1 | 127.0.0.1   | UDP      | 1232   | 63115 → 9999 Len=1200 |
| 28  | 0.336034 | 127.0.0.1 | 127.0.0.1   | UDP      | 1232   | 63115 → 9999 Len=1200 |
| 29  | 0.348577 | 127.0.0.1 | 127.0.0.1   | UDP      | 1232   | 63115 → 9999 Len=1200 |
| 30  | 0.361116 | 127.0.0.1 | 127.0.0.1   | UDP      | 1232   | 63115 → 9999 Len=1200 |

> Frame 1: Packet, 1232 bytes on wire (9856 bits), 1232 bytes captured (9856 bits) on interf

> Null/Loopback

> Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1

> User Datagram Protocol, Src Port: 63115, Dst Port: 9999

> Data (1200 bytes)

0000

02 00 00 00 45 00 04 cc 98 9b 00 00 40 11 00 00

0010

7f 00 00 01 7f 00 00 01 f6 8b 27 0f 04 b8 02 cc

0020

32 9e 00 0c 50 2b 4a eb 5d 9e 02 9a 08 0e 1c

0030

0e 82 98 a8 73 eb a5 7f d6 78 6d bb 50 be 4f 79

0040

65 e6 0d 03 7f ee e1 50 99 f4 f9 44 3b 5d c8 a7

0050

89 40 d7 2c 18 5d 4b 4e 97 ef 24 46 5a cc 69 7d

0060

ce fa 76 f6 44 28 0a 56 5f e6 fb 35 60 68 46 d2

0070

22 06 6e 5b c7 a7 b3 49 6f 89 b4 4f 2a be dc 50

0080

15 05 dc 86 6f 79 b7 5d 9f d1 09 da 37 5f f8 84

0090

ae d6 47 74 86 f0 52 4f d2 f4 db c7 83 f6 5e 7d

00a0

51 11 80 1e 2a b0 ae b1 94 61 09 75 05 99 74 e4

00b0

cd 1c 98 b4 2b af 0f 73 d1 3b dd 63 db bf f6 62

00c0

9e 37 ef 54 15 c1 94 1d d0 be 81 0e 95 76 2d b2

00d0

9e 64 25 26 12 93 4a 95 a7 71 d1 89 12 6c f0 4d

00e0

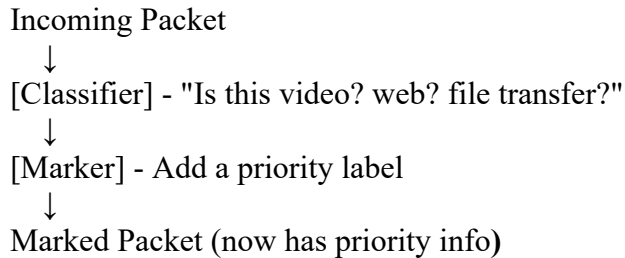
50 ed 17 77 bd f8 68 09 e8 4b 2f b2 a3 c1 e7 ed

00f0

4b c0 0e 74 c4 82 9b a3 55 27 dd 70 f5 51 6d 85

# QoS MECHANISMS

## 1. Classification and Marking: Identify and label packets based on their type.



### Common marking methods:

#### DSCP (Differentiated Services Code Point):

- 6-bit field in IP header
- Values: 0-63
- Examples:
  - DSCP 46 (EF) = Expedited Forwarding (highest priority - voice)
  - DSCP 34 (AF41) = Assured Forwarding (high priority - video)
  - DSCP 0 (BE) = Best Effort (lowest priority - bulk data)

#### 802.1p (Layer 2 marking):

- 3-bit field in Ethernet header
- Values: 0-7 (7 = highest priority)

## 2. Queueing Disciplines

- **FIFO (First-In-First-Out):**
  - Simplest: no prioritization
  - Not suitable for differentiated service
- **Priority Queuing:**
  - Multiple queues with strict priority
  - Risk: lower priority starvation
- **Weighted Fair Queuing (WFQ):**
  - Bandwidth sharing based on weights
  - Guarantees minimum bandwidth per class - Recommended for QUIC traffic

### 3. Congestion Avoidance

- **RED (Random Early Detection):**
  - Drops packets before queue is full
  - Prevents TCP global synchronization
  - Less relevant for QUIC (has own congestion control)
- **WRED (Weighted RED):**
  - Different drop thresholds per traffic class
  - Higher priority traffic dropped last

## POLICING MECHANISMS

| Feature | Policing                                           | Shaping                                        |
|---------|----------------------------------------------------|------------------------------------------------|
| Action  | Drops packets immediately if they exceed the rate. | Buffers packets in a queue to send them later. |
| Effect  | Causes instant packet loss; "choppy" traffic.      | Introduces Latency (Delay); "smooth" traffic.  |
| Visual  | The traffic graph looks "clipped" at the top.      | The traffic graph looks like a "smooth wave."  |
| Usage   | Common at ISP entry points (Ingress).              | Common at your router's exit (Egress).         |

#### 1) Token Bucket Algorithm:

##### Mechanism:

- Tokens generated at constant rate  $R$
- Bucket capacity  $B$  (burst size)
- Packet needs tokens equal to its size
- No tokens = packet dropped

##### Advantages:

- Allows controlled bursts
- Simple to implement
- Low computational overhead



## 2) Leaky Bucket Algorithm

### Mechanism:

- Packets enter at variable rate
- Leak out at constant rate  $R$
- Bucket capacity  $B$

### Characteristics:

- Smooths all traffic bursts
- Output always constant rate
- More restrictive than token bucket

### When to use:

- When constant output rate required
- Protecting downstream with strict rate limits
- Not ideal for QUIC (prevents legitimate bursts)

## 3) Dual Token Bucket (Two-Rate Policer)

### Mechanism:

- CIR bucket (Committed Information Rate)
- PIR bucket (Peak Information Rate)
- Three-color marking: GREEN, YELLOW, RED

### Marking Logic:

Packet size  $\leq$  CIR tokens  $\rightarrow$  GREEN (forward, high priority) CIR tokens  $<$  Packet  $\leq$  PIR  $\rightarrow$  YELLOW (forward, medium priority) Packet  $>$  PIR tokens  $\rightarrow$  RED (drop or lowest priority)

### Use Case:

- Enforce committed vs. burst rates
- Graceful degradation (YELLOW marked packets)
- More flexible than single token bucket

## 4) DiffServ Architecture

Most widely deployed QoS framework:

- Edge classification: Traffic identified and marked
- Core forwarding: Routers treat packets based on DSCP
- Scalable: No per-flow state in core